When lexicalized presupposition meets implicature: A causal, probabilistic model for *manage* and *fail*[1]

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Abstract. Recent work on factive verbs proposes that the not-at-issue inference associated with *know* can project through lexical encoding and pragmatic reasoning without a lexicalized presupposition (Scontras and Tonhauser, 2025). Since the not-at-issue content of implicative verbs also holds under both polarities, we ask whether this account extends to them. As our experiment shows, their not-at-issue inferences, as instantiated by attempt inferences from *manage* and *fail*, are strongly shaped by prior beliefs but show no QUD effect, and negative-outcome implicatives produce substantially stronger attempt inferences than outcome-matched baselines. Only three existing analyses, including two causal accounts (Baglini and Francez, 2016; Nadathur, 2023), are compatible with these findings; however, all leave the process of identifying the relevant background condition unspecified. We therefore develop a causally inspired Rational Speech Act account in which implicatives lexicalize the presupposition that some causally relevant condition be salient, and recursive pragmatic reasoning then combines this presupposition with contextual beliefs and alternative utterances to derive an attempt inference as an implicature, without requiring the identification of the causal condition. The model captures all the empirical findings and thus provides a unified probabilistic account of how attempt inferences arise, vary with context, and project.

Keywords: implicative verbs, attempt inferences, presupposition, projection, causal semantics, Rational Speech Act models

1. Introduction

A speaker who says *Alfred doesn't know that Catherine picked up the package* can deny Alfred's knowledge while conveying that Catherine picked it up. The latter inference survives negation and is therefore projective. Classical analyses derive factive projection from a lexically specified presupposition (Heim, 1983; van der Sandt, 1992), while reasoning-based analyses derive it from lexical meaning and conversational reasoning (e.g., Abusch, 2002; Roberts and Simons, 2024; Simons, 2001; Simons et al., 2017). Building on this work, Scontras and Tonhauser (2025) provide an explicit probabilistic implementation in the Rational Speech Act (RSA) framework. Their model derives the projection of the complement of *know* from lexical entailments, prior beliefs, the Question Under Discussion (QUD), and reasoning about the speaker's private assumptions, without specifying the complement as a lexical presupposition.

Implicative verbs such as *manage* and *fail* raise a related, but more complex, projection problem. *Alfred managed to hang up the photo* entails that he hung it up, whereas *Alfred did not manage to hang up the photo* entails that he did not. Yet both can suggest additional backgrounded content, such as effort or difficulty. Thus, negation reverses the complement outcome while some non-complement inferences may still project (Baglini and Francez, 2016; Karttunen, 1971; Karttunen and Peters, 1979; Nadathur, 2023; van Leusen, 2012). This motivates two related questions: what in the lexical meaning of implicative verbs gives rise to these non-complement,

[1]Experimental materials, data, statistical analyses, and RSA models are available at `https://github.com/alfreddongxuwu/manage-fail`.

not-at-issue inferences, and can their projection be derived through pragmatic reasoning without encoding them as lexical presuppositions, as proposed for *know*?[2]

Early analyses assign such inferences a purely conventional source: Karttunen (1971) treats trying as a presupposition of *manage*, while Karttunen and Peters (1979) analyze effort or difficulty as a conventional implicature. Subsequent analyses assign different roles to lexical content and contextual reasoning. Coleman (1975) proposes context-sensitive hierarchies of presuppositions; van Leusen (2012) combines a general implicative condition with accommodation and pragmatic strengthening; and causal accounts derive the inferential profile from backgrounded causal conditions (Baglini and Francez, 2016; Nadathur, 2023). These accounts disagree on which non-complement inferences are lexically encoded and how (or even whether) pragmatic reasoning contributes to their derivation, and their arguments rely largely on introspective judgments, with very limited empirical testing of their competing predictions.

This study therefore provides an experimental and computational account of how non-complement inferences associated with *manage* and *fail*, two canonical implicative verbs, arise, vary with context, and project. Since experimental testing requires a specific inference that can be manipulated and measured, we select attempt inferences because of their central role in the theoretical debate. These are inferences that the subject tried to bring about the event described by the complement. Inspired by Scontras and Tonhauser (2025), we conduct a controlled experiment manipulating prior beliefs and the QUD (Roberts, 2012). Three empirical findings emerge. First, participants infer an attempt more strongly when the preceding context makes trying more likely, consistent with previous findings that higher prior probability strengthens projection (Degen and Tonhauser, 2021; Scontras and Tonhauser, 2025). Second, the QUD manipulation has no reliable effect, despite previous evidence that content projects more strongly when it is not-at-issue (e.g., Tonhauser et al., 2018; Scontras and Tonhauser, 2025). Third, implicatives strengthen attempt inferences relative to outcome-matched baselines; we call this difference an implicative boost. Importantly, this boost comes mainly from negative-outcome forms: *failed* and *didn't manage* produce stronger attempt inferences than *didn't*. The positive-outcome forms, *managed* and *didn't fail*, only show a small boost over *did*.

A straightforward extension of Scontras and Tonhauser's (2025) approach that treats attempting as a lexical entailment of positive-polarity implicatives cannot explain why *managed* produces only slightly stronger attempt inferences than *did*. The results also challenge a fixed lexical requirement of attempting under both matrix polarities. Among the implicative analyses reviewed, only van Leusen (2012), Baglini and Francez (2016), and Nadathur (2023) remain compatible with all patterns. These three accounts appeal to world knowledge and contextual reasoning in identifying the relevant background condition needed to derive the attempt inference, but none explains the more specific process involved. Of these accounts, the latter two causal analyses provide a particularly concrete starting point: both encode a general causal condition on complement realization whose specific content is supposed to be supplied by context (Baglini and Francez, 2016; Nadathur, 2023). Building on this idea, we therefore develop a causally inspired

[2]With factive *know*, the complement is already entailed in positive sentences, so the central question concerns how that content projects from entailment-canceling environments such as negation and questions. For implicative verbs, the relevant non-complement inferences can vary even in positive sentences, making their source itself part of the explanatory problem. We therefore investigate both their derivation and their projection across unnegated and negated forms.

RSA model (Degen, 2023; Frank and Goodman, 2012) which represents this prerequisite separately from attempt and completion and captures all three empirical patterns, supporting the account that implicative verbs presuppose an actual, contextually salient causal condition with its interpretation left unspecified, and recursive pragmatic reasoning then combines this presupposition with contextual beliefs and alternative utterances to derive an attempt inference as an implicature.

Section 2 presents the experiment and evaluates existing analyses against the resulting empirical patterns. Section 3 develops the causal RSA model and compares it with two simpler models: one derives attempt inferences purely through pragmatic reasoning, while the other encodes them directly in the lexicon. Section 4 discusses the findings. Section 5 concludes.

2. Experiment

The main purpose of the experiment is to investigate what in the lexical meaning of *manage* and *fail* gives rise to attempt inferences, and whether pragmatic reasoning can derive their projection without encoding attempt as a lexical presupposition, as proposed for *know* by Scontras and Tonhauser (2025). For this account to hold, the following three predictions must be satisfied; we therefore test them experimentally.[3]

(1) a. **Prior effect.** Attempt inferences should be stronger when the preceding context makes an attempt more likely (Degen and Tonhauser, 2021; Scontras and Tonhauser, 2025).

b. **QUD effect.** Attempt inferences should be stronger when the QUD concerns completion of the event, leaving attempt not-at-issue, than when it directly concerns attempt (Tonhauser et al., 2018; Scontras and Tonhauser, 2025).

c. **Implicative boost.** Each implicative form should produce stronger attempt inferences than its outcome-matched baseline: *managed* and *didn't fail* compared with *did*, and *failed* and *didn't manage* compared with *didn't*.

2.1. Methods

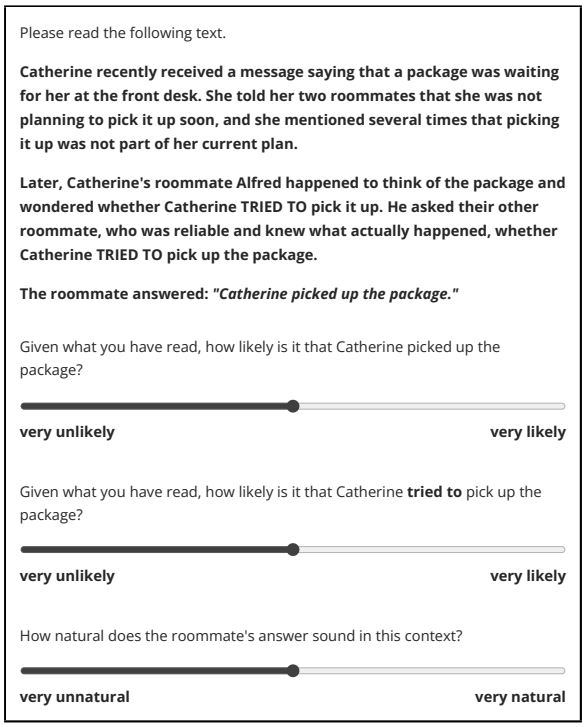
Materials and procedure. Participants read a short story ending with a roommate's reply, as illustrated in Figure 1.[4] Since previous research shows that *manage* can describe outcomes brought about without an attempt, as in "Harry managed to insult Ursula without even trying" (Nadathur 2023, example 15a), we selected two everyday intentional-action scenarios in which completing the action involves the subject actively trying to bring about the outcome: Alfred putting up a photo and Catherine picking up a package, a choice intended to minimize such "*managed without trying*" readings. Crossing the two items with prior belief (high vs. low), QUD (completion vs. attempt), and eight utterance types produced 64 conditions. Each participant completed one trial in a single condition.

[3]The first two predictions concern both unnegated and negated implicatives; in the latter, the attempt inference constitutes projection.

[4]The main experiment and norming study can be previewed at `https://alfreddongxuwu.github.io/manage-fail/main-experiment/` and `alfreddongxuwu.github.io/manage-fail/norming/`, respectively. Responses entered through these preview links are not submitted as study data.



(a) Item: photo; prior: high; QUD: P?; utterance: *didn't manage*.



(b) Item: package; prior: low; QUD: TRY?; utterance: *did*.

Figure 1: Sample trials illustrating the prior and QUD manipulations and the two orders of the likelihood questions.

The opening paragraph manipulated prior belief by describing the subject as planning or not planning to perform the action. For example, Alfred had repeatedly told his roommates that putting up the photo was, or was not, part of his current plan. These contexts established high and low prior probabilities of an attempt.[5]

The story then introduced the QUD. One roommate wondered whether the subject had completed the action (P?) or had tried to do so (TRY?), and asked another roommate who was described as reliable and knowledgeable about what had happened. In the TRY? condition, *tried to* was capitalized in both mentions of the question to make the QUD more salient and thereby strengthen its effect on interpretation. The reply took one of eight forms: *managed*, *didn't manage*, *failed*, *didn't fail*, *did*, *didn't*, *tried*, or *didn't try*. The *did* and *didn't* forms served as baselines for the implicatives. The *tried* and *didn't try* forms provided comparison points for the inference ratings.

Participants then rated the likelihood of an attempt, the likelihood of completion, and the naturalness of the reply. For the photo item, the attempt question asked, "Given what you have

[5]We independently normed the two items to verify the prior manipulation before introducing a QUD or target utterance. The norming study presented only the opening context and asked about attempt and completion in random order. Of 100 Prolific participants, 99 remained after excluding one who did not report English as a native language. Mean attempt ratings on the 0–1 scale were .817 (high) and .211 (low) for the photo item, and .861 (high) and .203 (low) for the package item. Thus, high-prior contexts elicited higher attempt ratings than low-prior contexts in both items, with similar ratings across items within each prior condition.

read, how likely is it that Alfred **tried to** put up the photo?" The completion question omitted *tried to*. Both used sliders from *very unlikely* to *very likely*. The naturalness slider ranged from *very unnatural* to *very natural*. The two likelihood questions appeared in random order, with naturalness always last. All three sliders had to be moved before participants could continue to a demographic questionnaire. Ratings were collected on a 0–100 scale and rescaled to 0–1 for analysis; higher attempt ratings indicate stronger attempt inferences.[6]

Participants. A total of 967 participants recruited through Prolific completed the experiment. We retained participants who reported English as a native language and residence in the United States, United Kingdom, Canada, Australia, or New Zealand. Applying these criteria excluded six participants, leaving a final sample of 961 (442 women, 511 men, 7 nonbinary participants, and 1 who declined to report gender; ages 18–76). Each trial took approximately five minutes, and participants were paid $1.

2.2. Results

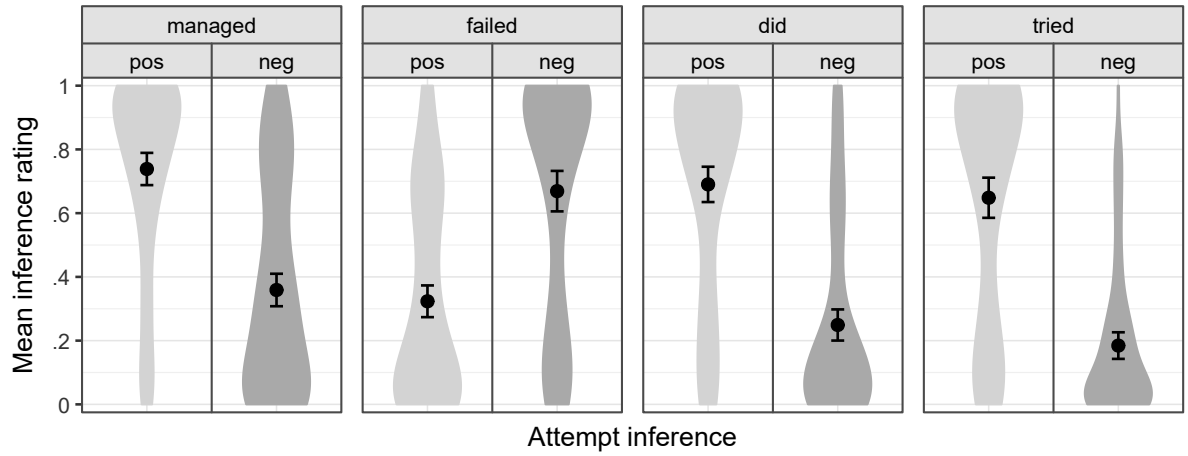


Figure 2: Distributions of attempt-inference ratings by utterance and matrix polarity (pos: unnegated; neg: negated), averaged across items, priors, and QUDs. Black points show means; error bars show 95% CIs.

Figure 2 shows gradient attempt inferences across the eight utterance types. To evaluate the three predictions, we fitted a linear mixed-effects model to attempt ratings from all eight utterance types, with fixed effects of utterance, prior, QUD, and rating-question order,[7] and a random intercept for item. Table 1 reports the three main comparisons, and Figure 3 displays the corresponding mean ratings.

[6] All descriptive means and RSA model inputs average participants within design cells and then weight the relevant cells equally. Main-experiment cells are defined by item, prior, QUD, and utterance; norming cells by item and prior. Question order is pooled within cells. RSA model predictions use the same condition weights.

[7] Scontras and Tonhauser (2025) used question order to reinforce the QUD manipulation by presenting the QUD-relevant rating question first. In our model, however, rating-question order had no significant main effect on attempt ratings ($\beta = -.004$, $p = .849$).

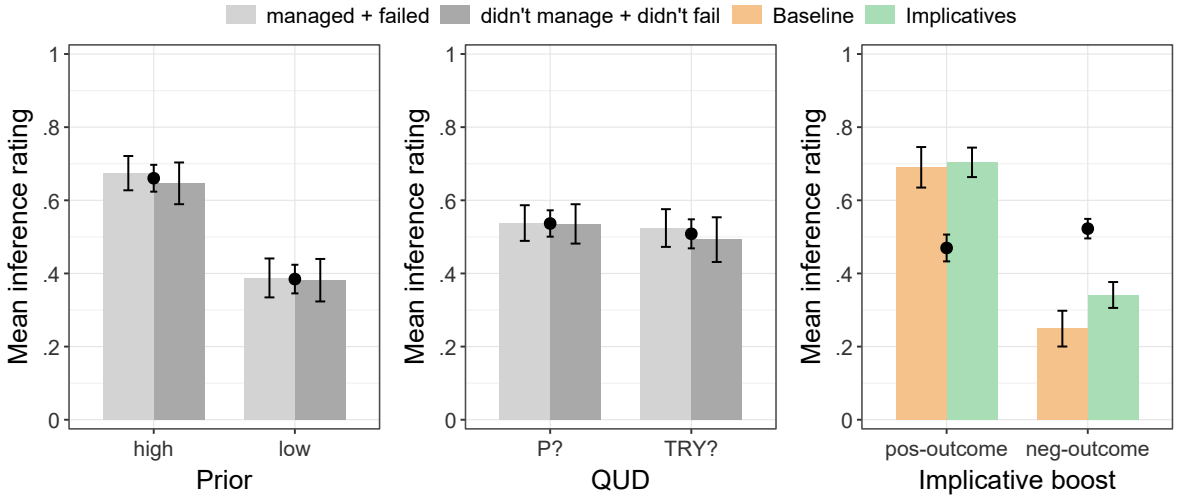


Figure 3: Attempt-inference ratings by prior, QUD, and implicative status. In the first two panels, bars separate matrix polarities and black points average all four implicatives. In the right panel, bars compare *managed* and *didn't fail* with *did* (positive outcome), and *didn't manage* and *failed* with *didn't* (negative outcome). The two black points in that panel show the overall baseline mean (left) and implicative mean (right), averaging across outcomes.

Comparison	β	SE	t	p
Prior: high − low	.253	.019	13.30	$< .001$
QUD: P? − TRY?	.012	.019	0.64	.521
Overall implicative boost	.054	.023	2.32	.021

Table 1: Model estimates for the three predictions. The overall boost compares the mean of the four implicatives with the mean of *did* and *didn't*. Estimates are on the 0–1 rating scale.

First, higher prior belief strengthened attempt inferences, supporting prediction (1a). Across the four implicatives, mean ratings increased from .385 in the low-prior condition to .660 in the high-prior condition (Figure 3, left). Second, the QUD manipulation did not significantly affect attempt inferences, so prediction (1b) was not supported. Across the four implicatives, mean ratings were .537 under P? and .508 under TRY? (Figure 3, middle), with a small, nonsignificant model estimate ($\beta = .012$, $p = .521$).

To examine projection specifically, we then fitted the same model to the two negated implicatives, testing whether prior belief and QUD modulated attempt inferences under negation. The same pattern held: higher prior belief strengthened projection ($\beta = .269$, $SE = .042$, $t(235) = 6.43$, $p < .001$), while the QUD manipulation had no significant effect ($\beta = .035$, $SE = .043$, $t(235) = 0.82$, $p = .415$).

Third, implicatives produced an overall boost relative to the baselines ($\beta = .054$, $p = .021$), providing partial support for prediction (1c) (Figure 3, right). The boost was concentrated in the negative-outcome forms: *didn't manage* and *failed* together exceeded *didn't* by .093 ($SE = .033$, $t(949) = 2.82$, $p = .005$), with both individual contrasts significant. By contrast, the

positive-outcome forms *managed* and *didn't fail* produced a small, nonsignificant boost over *did* ($\beta = .015$, $SE = .033$, $t(949) = 0.45$, $p = .653$).

2.3. Interim discussion

These findings separate two questions that are conflated in the factive case: where the non-complement inference comes from, and why it projects under negation. The positive-outcome conditions are informative about the first. Under a direct extension of Scontras and Tonhauser's account, if attempting were categorically encoded by *managed* in positive contexts, it should provide substantially stronger evidence for an attempt than its outcome-matched baseline *did*. Instead, the boost is quite small, arguing against a direct lexical encoding of attempting. The projection question is less decisive: the stronger attempt inference for *didn't manage* is compatible with projection, but the absence of a detectable QUD effect provides no independent support for extending the QUD-sensitive projection mechanism of Scontras and Tonhauser (2025) to this case. Their model nonetheless predicts the prior effect by combining prior beliefs with reasoning about the speaker's utterance choice. As this "projection without lexically-specified presupposition" analysis cannot account for all the empirical findings, the explanatory task is therefore to identify a lexical contribution from which contextual reasoning can derive attempt inferences across matrix polarities.

Existing analyses of implicative verbs offer several possible sources for these non-complement inferences. Since the role of QUD remains unresolved, we set it aside in the following comparisons and focus on two empirical targets. First, higher prior belief strengthens attempt inferences. Second, the implicative boost is concentrated in negative-outcome forms: *didn't manage* and *failed* produce a substantial boost over *didn't*, while *managed* and *didn't fail* produce only a small boost over *did*. Each outcome group contains an unnegated and a negated form, so matrix negation alone cannot explain this difference in boost.

Early analyses locate the additional content in conventional meaning. Karttunen (1971) assigns *manage* presuppositions involving difficulty and attempting within a more general analysis based on necessary and sufficient conditions for the complement. Karttunen and Peters (1979) analyze the effort or difficulty associated with *manage* as a conventional implicature. Their treatment of *fail* allows either trying or an expectation that the event would occur, with contextual plausibility determining which interpretation is supported. This contextual choice accommodates prior sensitivity. The conventional contribution itself, however, provides no distinction that explains why the attempt boost should be concentrated in negative-outcome conditions.

Coleman (1975) gives context a more explicit role through hierarchies of background inferences. For *manage*, an incompatible trying inference gives way to difficulty and then unlikelihood; *fail* has a separate hierarchy involving obligation, conditional necessity, and likelihood. Context thus determines which inference is available. The proposed ordering encounters cases where trying holds while difficulty is explicitly denied (Baglini and Francez, 2016), but the broader role assigned to context accommodates the prior effect. More recently, Knjižar (2025) treats effort as an optional component supplied through pragmatic enrichment, allowing it to vary with context and to be absent even with animate subjects. Both analyses accommodate prior sensitivity through contextual interpretation, while leaving the contrast between positive- and negative-outcome boosts unexplained.

Van Leusen (2012) provides a mechanism directly relevant to this contrast. On her account, *manage* introduces a general condition concerning sufficient effort, which context can specify as an intentional attempt. Unnegated *manage* permits outcomes achieved without trying. Negated *manage*, by contrast, supports an additional pragmatic inference: reasoning about alternatives produces the interpretation that some effort was made but was insufficient. This offers a route to an attempt inference from *didn't manage* that goes beyond the information conveyed by non-completion alone. Contextual specification accommodates the prior effect, and the additional inference from insufficient effort provides a qualitative explanation of the stronger negative-outcome boost.

Causal analyses offer a further explanation of why the outcome matters. Baglini and Francez (2016) propose that *manage p* presupposes a familiar catalyst that is causally necessary but insufficient for p, and asserts that this catalyst actually caused p. Trying is inferred when context identifies an intentional action as part of the catalyst. They explicitly observe that the catalyst requirement is satisfied when p is contingently true, whereas the falsity of p does not ensure its satisfaction; the presupposition is consequently more apparent under negation. This explains the relevance of the outcome-matched baselines: completing the action already supplies evidence of trying, while noncompletion leaves open whether an attempt occurred. The causal contribution can therefore add more evidence in negative-outcome conditions. Contextual identification of the catalyst accommodates prior sensitivity, while this difference in the information supplied by the outcome qualitatively matches the boost asymmetry.

Nadathur (2023) proposes that implicative verbs presuppose the existence of a causally necessary condition for the relevant outcome, with two-way implicatives additionally presupposing that it is the only unresolved such condition in the relevant background and hence causally sufficient. Unnegated implicative sentences assert that the condition is satisfied, while their negated counterparts assert that it is not. For *manage*, the relevant outcome is complement realization, and the prerequisite's content is left lexically unspecified. For polarity-reversing *fail*, the condition is necessary and sufficient for the complement's nonrealization. This analysis gives contextual causal knowledge a role in determining the relevant prerequisite and its relation to the outcome. Applied to our findings, it offers a basis for accommodating prior sensitivity and for allowing the additional evidence supplied by an implicative to depend on what is already established by completion or noncompletion.

As Table 2 shows, while most of the reviewed analyses cannot accommodate both findings, three remain compatible with both: van Leusen (2012), Baglini and Francez (2016), and Nadathur (2023). All three appeal to world knowledge and contextual reasoning to specify the relevant background conditions; however, none fully explains the specific process by which these conditions are identified. Van Leusen leaves the pragmatic rules governing accommodation unformalized; Baglini and Francez leave catalyst identification to an extra-grammatical process; and Nadathur treats causal models as contextual parameters, leaving the prerequisite associated with *manage* open to contextual enrichment. In the next section, we therefore build on the causal proposal that implicative verbs contribute a general, underspecified causal condition, and develop a simpler, probabilistic mechanism that accounts for both empirical findings without requiring the identification of that causal condition.

Analysis	Prior effect	Two boosts
Scontras and Tonhauser (2025)	✓	✗
Karttunen (1971)	✗	✗
Karttunen and Peters (1979)	✓	✗
Coleman (1975)	✓	✗
Knjižar (2025)	✓	✗
van Leusen (2012)	✓	✓
Baglini and Francez (2016)	✓	✓
Nadathur (2023)	✓	✓

Table 2: Compatibility of existing analyses with the empirical findings. ✓ indicates that the analysis accommodates or qualitatively predicts the finding; ✗ indicates that it does not.

3. Probabilistic modelling of attempt inferences

The Rational Speech Act (RSA) framework (Frank and Goodman, 2012; Degen, 2023) treats interpretation as probabilistic reasoning about a speaker's choice of utterance, combining lexical meaning, prior beliefs, and alternative descriptions. In this section, we use the RSA framework to model the derivation of attempt inferences, including their projection under negation. We first compare two baseline models that differ in whether implicatives lexically require an attempt, then develop a causally inspired model. Finally, we examine two extensions: mapping predicted attempt probabilities to ratings for each utterance and incorporating QUD.

3.1. Shared assumptions and model settings

Variables and alternatives. All models represent actual attempting with T and complement completion with O. Both are binary; c identifies the prior context:

$$T, O \in \{0, 1\}, \qquad c \in \{\text{high}, \text{low}\}.$$

The eight experimental utterances form the common alternative set:

$$\mathcal{U} = \left\{ \begin{array}{llll} \textit{managed}, & \textit{didn't manage}, & \textit{failed}, & \textit{didn't fail}, \\ \textit{did}, & \textit{didn't}, & \textit{tried}, & \textit{didn't try} \end{array} \right\}.$$

Shared priors. The norming study supplies the attempt and completion probabilities in Table 3, averaging the two items equally. To obtain a joint distribution over T and O, we assume that completion probabilities conditional on attempt status are constant across contexts. Let $r_1 = P(O = 1 \mid T = 1)$ and $r_0 = P(O = 1 \mid T = 0)$. Solving

$$P(O = 1 \mid c) = r_1 P(T = 1 \mid c) + r_0[1 - P(T = 1 \mid c)] \tag{1}$$

for the high and low contexts gives $r_1 = .944$ and $r_0 = .055$. The joint prior is therefore $P(T, O \mid c) = P(T \mid c)P(O \mid T)$. All models use this empirical distribution; the causal model extends it with a variable for causal-condition identity.

Prior context	$P(T = 1 \mid c)$	$P(O = 1 \mid c)$
High	.839	.801
Low	.207	.239

Table 3: Norming-based attempt and completion priors.

Utterance costs and speaker optimality. The shared cost function is $\text{Cost}(u) = -\log N(u)$, where $N(u)$ is the mean naturalness rating for utterance u in the main experiment. More natural utterances thus have lower costs. These naturalness ratings by utterance are reported in the Appendix (Table 15). Costs are fixed across contexts, and speaker optimality is fixed at $\alpha = 1$ throughout.[8]

RSA inference. Let $[[u]](w)$ be 1 when world w satisfies the semantic requirements of utterance u, and 0 otherwise. Let $\pi_c(w)$ denote the contextual world prior. The literal listener updates this prior with the utterance's meaning:

$$L_0(w \mid u, c) \propto [[u]](w)\pi_c(w). \tag{2}$$

The speaker chooses among the alternatives in $\mathcal{U}$ according to their informativeness for this listener and their cost:

$$S_1(u \mid w, c) \propto \exp\{\alpha[\log L_0(w \mid u, c) - \text{Cost}(u)]\}. \tag{3}$$

The pragmatic listener then infers the world from the speaker's choice:

$$L_1(w \mid u, c) \propto S_1(u \mid w, c)\pi_c(w). \tag{4}$$

The two baseline models and the causal model share this recursion. The backoff extension changes the final response mapping, and the QUD extension changes the speaker's information goal.

Predictions and comparison targets. The predicted attempt rating is the pragmatic listener's probability that an attempt occurred:

$$\widehat{y}(u, c) = \sum_{w:T(w)=1} L_1(w \mid u, c). \tag{5}$$

Our central target is the additional inference contributed by an implicative beyond its complement outcome baseline. We therefore evaluate predictions through outcome-matched contrasts:

$$\Delta(u, c) = \widehat{y}(u, c) - \widehat{y}(b(u), c), \tag{6}$$

where $b(u)$ is *did* for *managed* and *didn't fail*, and *didn't* for *failed* and *didn't manage*. Averaging the two contrasts within each outcome group gives the positive- and negative-outcome boosts. We also compare the high-minus-low prior difference across the four implicatives.

The observed values used for model comparison are descriptive contrasts; the inferential estimates are reported in Section 2. The Appendix gives the predictions for every utterance in all five models.

[8] Among the values examined in our sensitivity analysis (1–10), $\alpha = 1$ provides the closest fit to the empirical findings for the causal model. We therefore use this value across all models to facilitate comparison.

3.2. Two baseline models

The two baseline models represent opposing assumptions about the lexical source of attempt inferences. Motivated by analyses that conventionally associate implicatives with trying or effort (Karttunen, 1971; Karttunen and Peters, 1979), *M-attempt* tests a uniform lexical requirement that an attempt occurred. At the other endpoint, *M-outcome* assigns implicatives only their complement-outcome meanings and tests how much can be derived through prior beliefs and pragmatic reasoning.

Worlds and meanings. Both models use the same four worlds, with $w_{to} = \langle t, o \rangle$:

$$\mathcal{W}_{\text{baseline}} = \{w_{11}, w_{10}, w_{01}, w_{00}\}.$$

Table 4 gives their contextual priors. Table 5 places the two lexicons side by side. *M-attempt* adds $T = 1$ to the four implicative meanings, including the negated forms; the other utterances retain their meanings.

World	$\langle T, O \rangle$	High prior	Low prior
w_{11}	$\langle 1, 1 \rangle$	.792	.195
w_{10}	$\langle 1, 0 \rangle$	.047	.012
w_{01}	$\langle 0, 1 \rangle$	.009	.043
w_{00}	$\langle 0, 0 \rangle$	.152	.750

Table 4: Worlds and prior probabilities shared by *M-outcome* and *M-attempt*.

	M-outcome		*M-attempt*	
Utterance	Condition	Worlds	Condition	Worlds
managed	$O = 1$	$\{w_{11}, w_{01}\}$	$T = 1 \wedge O = 1$	$\{w_{11}\}$
didn't manage	$O = 0$	$\{w_{10}, w_{00}\}$	$T = 1 \wedge O = 0$	$\{w_{10}\}$
failed	$O = 0$	$\{w_{10}, w_{00}\}$	$T = 1 \wedge O = 0$	$\{w_{10}\}$
didn't fail	$O = 1$	$\{w_{11}, w_{01}\}$	$T = 1 \wedge O = 1$	$\{w_{11}\}$
did	$O = 1$	$\{w_{11}, w_{01}\}$	$O = 1$	$\{w_{11}, w_{01}\}$
didn't	$O = 0$	$\{w_{10}, w_{00}\}$	$O = 0$	$\{w_{10}, w_{00}\}$
tried	$T = 1$	$\{w_{11}, w_{10}\}$	$T = 1$	$\{w_{11}, w_{10}\}$
didn't try	$T = 0$	$\{w_{01}, w_{00}\}$	$T = 0$	$\{w_{01}, w_{00}\}$

Table 5: Semantic conditions and compatible worlds for the two baseline models.

Predictions. *M-outcome* produces a prior effect of .251 but zero implicative boost in both outcome groups. Each implicative has the same meaning as its outcome-matched baseline and consequently produces the same attempt inference. *M-attempt*, in turn, predicts an attempt probability of 1 for every implicative in both prior contexts. It therefore predicts a prior effect of zero and positive- and negative-outcome boosts of .196 and .957, respectively. Table 9 compares these predictions with the experimental data. For predictions by utterance, see the Appendix (Tables 10 and 11). These results motivate a lexical contribution that can strengthen attempt inferences relative to outcome-matched baselines while allowing their strength to vary with context.

3.3. A causally inspired model

Building on the causal analyses discussed in Section 2.3, *M-causal* assigns implicatives a general causal condition whose specific identity need not be resolved.

Worlds and causal conditions. The model extends the baseline representation with a variable C, which records whether a causal condition is salient and, if so, whether an attempt or another condition supplies it:

$$w = \langle T, O, C \rangle, \qquad C \in \{\text{TRY}, \text{OTHER}, \text{NONE}\}, \qquad C = \text{TRY} \implies T = 1.$$

With C = TRY, an actual attempt supplies the salient causal condition; with C = OTHER, another enabling or blocking condition supplies it. With C = NONE, no such condition is salient for the implicative description. An attempt can supply the condition only if it occurred. This leaves the ten worlds listed in Table 7.

Causal priors. The world prior combines the shared attempt–outcome distribution with a distribution over causal-condition identity:

$$\pi_c(T, O, C) = P(T \mid c)P(O \mid T)P(C \mid T, O). \tag{7}$$

Table 6 specifies $P(C \mid T, O)$ and provides an intuitive probabilistic structure that relates causal salience to attempting and completion. When an attempt occurred, we distribute probability equally across the three possibilities for causal salience, leaving multiple causal interpretations available. Completion without an attempt supports an alternative causal route, so we assign greater probability to OTHER. When neither attempting nor completion occurred, we assign greater probability to NONE. The resulting world priors (see Table 7) are then combined with utterances through recursive reasoning to probabilistically derive attempt inferences.

T	O	C = TRY	C = OTHER	C = NONE
1	1	1/3	1/3	1/3
1	0	1/3	1/3	1/3
0	1	0	2/3	1/3
0	0	0	1/3	2/3

Table 6: The conditional distribution $P(C \mid T, O)$ in *M-causal*.

World	$\langle T, O, C \rangle$	High prior	Low prior
v_1	$\langle 1, 1, \text{TRY} \rangle$	.264	.065
v_2	$\langle 1, 1, \text{OTHER} \rangle$	.264	.065
v_3	$\langle 1, 1, \text{NONE} \rangle$	.264	.065
v_4	$\langle 1, 0, \text{TRY} \rangle$	.016	.004
v_5	$\langle 1, 0, \text{OTHER} \rangle$	.016	.004
v_6	$\langle 1, 0, \text{NONE} \rangle$	.016	.004
v_7	$\langle 0, 1, \text{OTHER} \rangle$	.006	.029
v_8	$\langle 0, 1, \text{NONE} \rangle$	.003	.014
v_9	$\langle 0, 0, \text{OTHER} \rangle$	.051	.250
v_{10}	$\langle 0, 0, \text{NONE} \rangle$	.101	.500

Table 7: The ten worlds and their prior probabilities in *M-causal*.

Lexical meanings. All four implicatives presuppose $C \neq$ NONE and retain their outcome requirements. Table 8 gives the resulting meanings and compatible worlds. The causal requirement holds across matrix polarities. The predicted probability of an attempt is obtained by summing over all compatible worlds with $T = 1$, including worlds with $C =$ OTHER.

Utterance	Condition	Compatible worlds
managed	$O = 1 \wedge C \neq$ NONE	$\{v_1, v_2, v_7\}$
didn't manage	$O = 0 \wedge C \neq$ NONE	$\{v_4, v_5, v_9\}$
failed	$O = 0 \wedge C \neq$ NONE	$\{v_4, v_5, v_9\}$
didn't fail	$O = 1 \wedge C \neq$ NONE	$\{v_1, v_2, v_7\}$
did	$O = 1$	$\{v_1, v_2, v_3, v_7, v_8\}$
didn't	$O = 0$	$\{v_4, v_5, v_6, v_9, v_{10}\}$
tried	$T = 1$	$\{v_1, v_2, v_3, v_4, v_5, v_6\}$
didn't try	$T = 0$	$\{v_7, v_8, v_9, v_{10}\}$

Table 8: Semantic conditions and compatible worlds for *M-causal*.

Predictions. As Table 9 shows, *M-causal* successfully captures the prior effect, the small positive-outcome boost, and especially the larger negative-outcome boost. For predictions by utterance, see the Appendix (Table 12). Section 4 presents a more detailed discussion of these results and how *M-causal* derives them.

Model	Prior effect	Positive boost	Negative boost
M-outcome	.251	.000	.000
M-attempt	.000	.196	.957
M-causal	**.306**	**.011**	**.092**
Observed	**.276**	**.014**	**.092**

Table 9: Predicted and observed contrasts on the 0–1 attempt-rating scale. The prior effect compares high and low contexts across the four implicatives; boosts compare implicatives with their outcome-matched baselines.

3.4. Two extensions

We next examine two extensions of *M-causal*. The first addresses the model's tendency to predict more extreme attempt ratings than participants give; the second introduces the experimental QUD into the speaker's information goal.

3.4.1. Backoff to contextual priors

Although *M-causal* captures the contrasts, its predictions for individual utterances can be more extreme than participants' ratings. For example, it assigns *tried* an attempt probability of 1 and *didn't try* a probability of 0. Following the response treatment of Scontras and Tonhauser (2025), *M-backoff* allows listeners to rely more on their contextual prior when the interpretation of an utterance departs strongly from that prior.

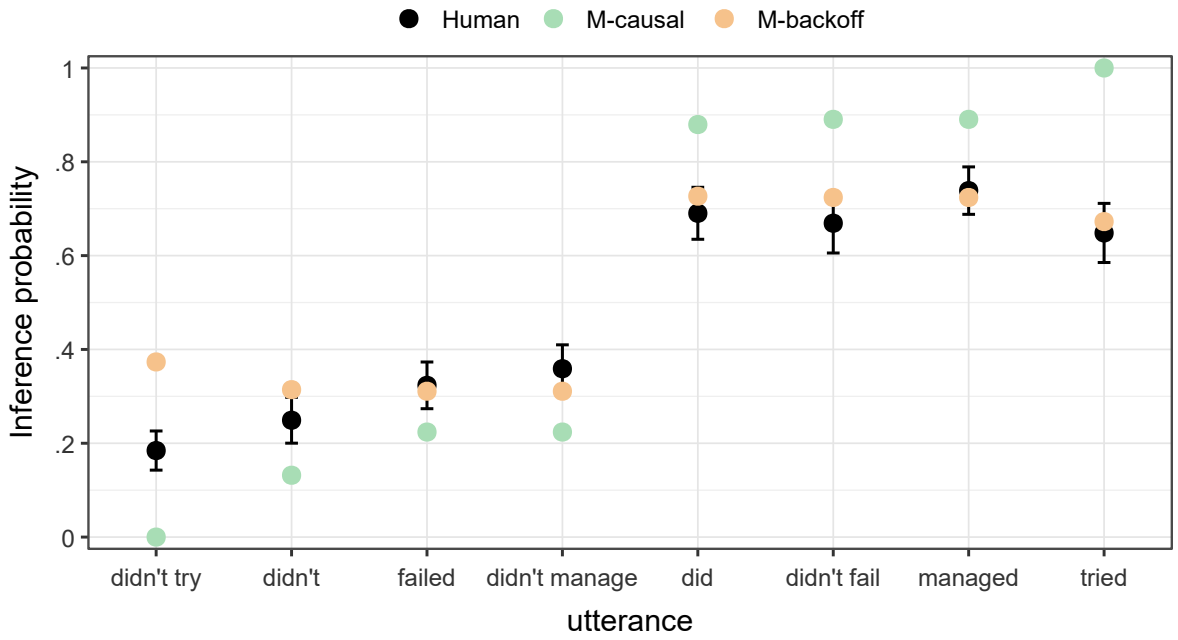


Figure 4: Attempt-inference ratings and predictions from *M-causal* and *M-backoff*. Black points show experimental means, with 95% CIs. Colored points show model predictions.

Let $p_{u,c}$ be the attempt probability predicted by *M-causal* and $q_c = P(T = 1 \mid c)$ the contextual prior. Their Kullback–Leibler divergence is $D_{u,c} = D_{\mathrm{KL}}(\mathrm{Bern}(p_{u,c}) \parallel \mathrm{Bern}(q_c))$, where $\mathrm{Bern}(p)$ is a Bernoulli distribution with attempt probability p. The predicted response is

$$\widehat{y}_{\text{backoff}}(u, c) = e^{-D_{u,c}} p_{u,c} + (1 - e^{-D_{u,c}}) q_c. \tag{8}$$

Larger divergence places more weight on the prior.

This response mapping brings the overall predictions for the eight utterances closer to the observed means (Figure 4; see also the Appendix, Table 13). However, it also reduces the negative-outcome boost from .092 to −.003 and increases the prior effect from .306 to .511. Pulling both implicatives and baselines toward the same contextual prior removes the difference that defines the negative boost. Backoff therefore illustrates why that contrast is an essential modelling target.

3.4.2. QUD-based informativeness

The second extension asks how attempt inferences change when the speaker aims to resolve the experimentally supplied QUD. Following the partition-based treatment of relevance in RSA (Degen, 2023), *M-QUD* groups worlds according to whether the subject tried, $Q_{\mathrm{TRY}}(w) = T(w)$, or completed the action, $Q_{\mathrm{P}}(w) = O(w)$. The literal listener's probability of the correct answer is the total probability of worlds giving that answer:

$$L_0(Q(w) \mid u, c) = \sum_{w' : Q(w') = Q(w)} L_0(w' \mid u, c). \tag{9}$$

The speaker chooses among utterances compatible with the actual world, substituting this answer probability for the world probability in (3). The pragmatic listener then interprets the

utterance given the observed QUD. Worlds, priors, lexical meanings, costs, and the response mapping are otherwise shared with *M-causal*.

The QUD extension preserves the main findings: the prior effect is .272, the positive-outcome boost is .004, and the negative-outcome boost is .113. Across the four implicatives, the predicted QUD difference, P? minus TRY?, is only −.009. Averaging just the two negated forms, *didn't manage* and *didn't fail*, produces the same difference of −.009. For predictions by utterance under each QUD, see the Appendix (Table 14).

Thus, the model predicts little overall QUD effect on attempt inferences, including when the analysis is restricted to projection under negation. This parallels the experimental findings (Section 2) and seems to support the view that attempt projection from *manage* and *fail* is an exception to the Gradient Projection Principle (Tonhauser et al., 2018).

4. Discussion

The three main models show what is needed to explain the prior effect and the implicative boost together. *M-outcome* captures prior sensitivity, but its outcome-only meanings make implicatives equivalent to their outcome-matched baselines and produce no boost. *M-attempt* requires an attempt for every implicative, eliminating prior sensitivity and predicting boosts much larger than those observed. *M-causal* captures both findings by combining a general causal presupposition with a probabilistic structure that relates causal salience to attempting and completion. Recursive reasoning combines contextual world priors built on this structure with the speaker's utterance.

Prior effect and implicative boost. The listener infers a probability distribution over worlds from the speaker's utterance, then sums the probabilities of worlds in which an attempt occurred. If the context makes attempting likely, these worlds receive greater prior probability. If attempting is unlikely, worlds without an attempt receive greater prior probability, and another condition can still support the implicative description. The same lexical requirement therefore produces attempt inferences of different strengths across prior contexts. This account links prior effects on projection (Degen and Tonhauser, 2021; Scontras and Tonhauser, 2025) to the causal contribution of implicatives.

The outcome determines how much information the causal requirement adds. Noncompletion leaves open whether an attempt occurred. In the model's negative-outcome worlds, a salient causal condition is twice as probable when the subject tried as when they did not. Hearing *failed* or *didn't manage* therefore supplies additional evidence for an attempt beyond hearing *didn't*. Given this probabilistic structure, the causal requirement produces a negative-outcome boost of .080 at L_0. Recursive reasoning about utterance choice increases this boost to .092 at L_1, accurately matching the observed boost of .092.

Completion already provides strong evidence for attempting. Moreover, in positive-outcome worlds, a salient causal condition has probability $2/3$ both with and without an attempt. The causal requirement therefore adds no evidence for attempting at the literal level: *managed* and *didn't fail* produce the same attempt probability as *did*, showing a positive-outcome boost of zero at L_0. The listener nevertheless reasons about the speaker's choice among the implicative, the

outcome description, and the explicit alternatives *tried* and *didn't try*. This reasoning slightly shifts the balance between attempting and nonattempting worlds, producing a small average positive-outcome boost of .011, close to the observed .014.

The same mechanism also explains attempt inferences under negation. Matrix negation reverses the complement outcome while preserving the general causal requirement, which supports the inference in both *didn't manage* and *didn't fail*. These results therefore support the joint contribution of the causal presupposition, the probabilistic structure, and pragmatic reasoning.

Attempt inference without condition identification. The causal analyses discussed in Section 2.3 motivate a context-sensitive causal contribution while leaving open how listeners identify its content; our model derives attempt inferences without requiring this identification. The intuitively motivated distribution $P(C \mid T, O)$ relates causal salience to attempting and completion. Combined with the independently normed attempt and completion probabilities, this structure determines the contextual world priors. The RSA listener combines these priors with the utterance through recursive reasoning, then sums over all worlds in which an attempt occurred. This includes worlds in which another condition supplies the salient causal contribution. The attempt inference can therefore be derived while the identity of the causal condition remains unresolved.

The role of event knowledge in this reasoning also connects to Roberts and Simons's (2024) account of projection, as they explain how listeners use the semantic properties of events and world knowledge to infer background assumptions that make an utterance informative.

QUD and projection. The experiment found no reliable overall QUD effect across the four implicatives or within the two negated forms, and the QUD-based model likewise predicts only a small aggregate effect. This pattern connects to Degen and Tonhauser's (2025) finding that at-issueness effects vary with lexical meaning, extending the question of QUD sensitivity to non-complement inferences supported by a causal requirement and suggesting that implicative attempt projection may depart from the Gradient Projection Principle (Tonhauser et al., 2018).

Another possibility concerns how listeners interpret the experimental QUD. Scontras and Tonhauser (2025) suggest that an insufficiently clear QUD manipulation may explain why experimental QUD effects are weaker than their model predicts. Their later model (Scontras and Tonhauser, 2026) allows listeners to infer the operative QUD from the utterance, so that different presented questions can produce similar projection inferences. Extending this approach to implicatives would allow us to examine how inference about the QUD affects the derivation of attempt inferences.

Lexicalized presupposition meets implicature. The results therefore support a joint contribution of lexicalized presupposition and implicature. The use of an implicative presupposes that some contextually salient causal condition relevant to the outcome holds, and pragmatic reasoning about the utterance and contextual world priors then derives the attempt inference while leaving the causal condition unspecified. A high prior strengthens the attempt inference, and in negative-outcome cases the causal requirement adds evidence for attempting beyond noncompletion alone. Because this requirement holds across matrix polarities, the same reasoning also derives the projection of attempt inferences under negation.

5. Conclusion

We show that attempt inferences from *manage* and *fail* are sensitive to prior beliefs and receive a larger implicative boost in negative-outcome forms, while QUD manipulation produces no reliable overall effect. These findings cannot be explained by a direct extension of Scontras and Tonhauser's (2025) account or by treating attempting as a fixed lexical presupposition, but favor contextual analyses of background conditions (van Leusen, 2012; Baglini and Francez, 2016; Nadathur, 2023). As these analyses leave the process of identifying the relevant condition unspecified, our causally inspired RSA model derives attempt inferences without requiring this identification. It specifies an intuitive probabilistic structure that relates causal salience to attempting and completion and incorporates that structure into contextual world priors. Recursive reasoning combines these priors with the utterance and its alternatives to derive attempt inferences and their projection under negation. Through the interaction of a lexicalized causal presupposition and a context-sensitive attempt implicature, the model captures both prior sensitivity and the contrast between small positive-outcome and larger negative-outcome boosts.

This study therefore demonstrates how context-sensitive accounts of projection can be extended to the inferential behavior of implicative verbs by combining them with causal semantics within the RSA framework. Future work can further investigate whether and under what conditions QUD effects emerge and examine other types of not-at-issue inferences across a broader range of implicative verbs.

References

Abusch, D. (2002). Lexical alternatives as a source of pragmatic presuppositions. In B. Jackson (Ed.), *Proceedings of Semantics and Linguistic Theory 12*, Ithaca, NY, pp. 1–19. CLC Publications.

Baglini, R. and I. Francez (2016). The implications of managing. *Journal of Semantics 33*(3), 541–560.

Coleman, L. (1975). The case of the vanishing presupposition. In C. Cogen, H. Thompson, G. Thurgood, K. Whistler, and J. Wright (Eds.), *Proceedings of the First Annual Meeting of the Berkeley Linguistics Society*, Berkeley, CA, pp. 78–89. Berkeley Linguistics Society.

Degen, J. (2023). The Rational Speech Act framework. *Annual Review of Linguistics 9*, 519–540.

Degen, J. and J. Tonhauser (2021). Prior beliefs modulate projection. *Open Mind 5*, 59–70.

Degen, J. and J. Tonhauser (2025). Projection inferences: On the relation between prior beliefs, at-issueness, and lexical meaning. *Glossa Psycholinguistics 4*(1), 1–34. Article 19.

Frank, M. C. and N. D. Goodman (2012). Predicting pragmatic reasoning in language games. *Science 336*(6084), 998.

Heim, I. (1983). On the projection problem for presuppositions. In M. Barlow, D. Flickinger, and M. T. Wescoat (Eds.), *Proceedings of the Second West Coast Conference on Formal Linguistics*, Stanford, CA, pp. 114–125. Stanford Linguistics Association.

Karttunen, L. (1971). Implicative verbs. *Language 47*(2), 340–358.

Karttunen, L. and S. Peters (1979). Conventional implicature. In C.-K. Oh and D. A. Dinneen (Eds.), *Presupposition*, Volume 11 of *Syntax and Semantics*, pp. 1–56. New York, NY: Academic Press.

Knjižar, I. Z. (2025). The mechanisms of interpreting the implicative verb *manage*: A relevance-

theoretic perspective. *Annals of the Faculty of Philology 37*(2), 51–66.

Nadathur, P. (2023). Causal semantics for implicative verbs. *Journal of Semantics 40*(2–3), 311–358.

Roberts, C. (2012). Information structure in discourse: Towards an integrated formal theory of pragmatics. *Semantics and Pragmatics 5*, 1–69. Article 6.

Roberts, C. and M. Simons (2024). Preconditions and projection: Explaining non-anaphoric presupposition. *Linguistics and Philosophy 47*(4), 703–748.

Scontras, G. and J. Tonhauser (2025). Projection without lexically-specified presupposition: A model for *know*. In F. Longo and D. Panizza (Eds.), *Proceedings of Sinn und Bedeutung 29*, Noto, pp. 1431–1448. University of Messina, C.U.M.O.

Scontras, G. and J. Tonhauser (2026). Predicting within-utterance projection variation. *Meaning: A Journal of Linguistics and Philosophy 2*, 1–27. Article 2.

Simons, M. (2001). On the conversational basis of some presuppositions. In R. Hastings, B. Jackson, and Z. Zvolenszky (Eds.), *Proceedings of Semantics and Linguistic Theory 11*, Ithaca, NY, pp. 431–448. CLC Publications.

Simons, M., D. Beaver, C. Roberts, and J. Tonhauser (2017). The best question: Explaining the projection behavior of factives. *Discourse Processes 54*(3), 187–206.

Tonhauser, J., D. I. Beaver, and J. Degen (2018). How projective is projective content? Gradience in projectivity and at-issueness. *Journal of Semantics 35*(3), 495–542.

van der Sandt, R. A. (1992). Presupposition projection as anaphora resolution. *Journal of Semantics 9*(4), 333–377.

van Leusen, N. (2012). The accommodation potential of implicative verbs. In M. Aloni, V. Kimmelman, F. Roelofsen, G. W. Sassoon, K. Schulz, and M. Westera (Eds.), *Logic, Language and Meaning*, Volume 7218 of *Lecture Notes in Computer Science*, Berlin, Heidelberg, pp. 421–430. Springer.

Appendix

Utterance	High	Low	Overall	Observed
managed	.995	.779	.887	.739
didn't manage	.295	.009	.152	.359
failed	.295	.009	.152	.323
didn't fail	.995	.779	.887	.669
did	.995	.779	.887	.690
didn't	.295	.009	.152	.249
tried	1.000	1.000	1.000	.648
didn't try	.000	.000	.000	.184

Table 10: Final predictions from *M-outcome*.

Utterance	High	Low	Overall	Observed
managed	1.000	1.000	1.000	.739
didn't manage	1.000	1.000	1.000	.359
failed	1.000	1.000	1.000	.323
didn't fail	1.000	1.000	1.000	.669
did	.994	.613	.804	.690
didn't	.086	.000	.043	.249
tried	1.000	1.000	1.000	.648
didn't try	.000	.000	.000	.184

Table 11: Final predictions from *M-attempt*.

Utterance	High	Low	Overall	Observed
managed	.994	.787	.890	.739
didn't manage	.426	.022	.224	.359
failed	.426	.022	.224	.323
didn't fail	.994	.787	.890	.669
did	.995	.763	.879	.690
didn't	.258	.006	.132	.249
tried	1.000	1.000	1.000	.648
didn't try	.000	.000	.000	.184

Table 12: Final predictions from *M-causal*.

Utterance	High	Low	Overall	Observed
managed	.973	.475	.724	.739
didn't manage	.573	.049	.311	.359
failed	.573	.049	.311	.323
didn't fail	.973	.475	.724	.669
did	.973	.480	.727	.690
didn't	.586	.043	.314	.249
tried	.974	.371	.673	.648
didn't try	.704	.043	.373	.184

Table 13: Final predictions from *M-backoff*.

	P?			TRY?		
Utterance	High	Low	Overall	High	Low	Overall
managed	.985	.773	.879	1.000	.908	.954
didn't manage	.451	.039	.245	.375	.003	.189
failed	.451	.039	.245	.375	.003	.189
didn't fail	.985	.773	.879	1.000	.908	.954
did	.983	.748	.866	1.000	.918	.959
didn't	.283	.019	.151	.115	.001	.058
tried	1.000	1.000	1.000	1.000	1.000	1.000
didn't try	.000	.000	.000	.000	.000	.000

Table 14: Final predictions from *M-QUD* under each experimental QUD.

Utterance	Mean naturalness
managed	.608
didn't manage	.527
failed	.365
didn't fail	.230
did	.630
didn't	.670
tried	.554
didn't try	.589

Table 15: Mean naturalness ratings by utterance in the main experiment, on the 0–1 scale.